4.8.3

since the distribution is normal, we know that

$$f_k(x) = \frac{1}{\sqrt{2\pi}\,\sigma_k} e^{\left(-\frac{1}{2\sigma_k^2}(x-\mu_k)^2\right)}$$

and plugging $f_k(x)$ into Bayes' Theorem gives

$$p_k(x) = \frac{\pi_k \frac{1}{\sqrt{2\pi}\,\sigma_k} e^{\left(-\frac{1}{2\sigma_k^2}(x-\mu_k)^2\right)}}{\sum_{l=1}^{K} \pi_l \frac{1}{\sqrt{2\pi}\,\sigma_l} e^{\left(-\frac{1}{2\sigma_l^2}(x-\mu_l)^2\right)}}$$

we want to maximize $p_k(x)$, which is equivalent to maximizing its numerator since the denominator is constant for all values of k. it is further equivalent to maximizing the ln of the numerator since logarithms are monotonically increasing. therefore we can write the discriminant function as

$$\delta_k(x) = \ln(\pi_k) + \ln\left(\frac{1}{\sqrt{2\pi}}\right) - \ln(\sigma_k) - \frac{(x-\mu_k)^2}{2\sigma_k^2}$$

we drop $\ln\left(\frac{1}{\sqrt{2\pi}}\right)$ and expand to rewrite

$$\delta_k(x) = \ln(\pi_k) - \ln(\sigma_k) - \frac{x^2}{2\sigma_k^2} + \frac{x\mu_k}{\sigma_k^2} - \frac{\mu_k^2}{2\sigma_k^2}$$

since σ_k^2 is <u>not</u> constant across classes, we cannot drop the quadratic term $-\frac{x^2}{2\sigma_k^2}$. therefore $\delta_k(x)$ is quadratic and not linear.

4.8.7

using Bayes' Theorem:

$$P(Y = \text{yes} \mid X = 4) = \frac{\pi_{\text{yes}} f_{\text{yes}}(4)}{\pi_{\text{yes}} f_{\text{yes}}(4) + \pi_{\text{no}} f_{\text{no}}(4)}$$

using the normal density function,

$$f_{\text{yes}}(4) = \frac{1}{\sqrt{2\pi}\,6} e^{-\frac{1}{72}(4-10)^2}$$

$$= \frac{1}{6\sqrt{2\pi}} e^{-\frac{1}{2}} = \frac{1}{6\sqrt{2\pi e}}$$

$$f_{\text{no}}(4) = \frac{1}{\sqrt{2\pi}\,6} e^{-\frac{1}{72}(4-0)^2}$$

$$= \frac{1}{6\sqrt{2\pi}} e^{-\frac{2}{9}}$$

furthermore, $\pi_{\text{yes}} = 0.8$ and $\pi_{\text{no}} = 0.2$. then

$$P(Y = \text{yes} \mid X = 4) = \frac{0.8 \cdot \frac{1}{6\sqrt{2\pi e}}}{0.8 \cdot \frac{1}{6\sqrt{2\pi e}} + 0.2 \cdot \frac{1}{6\sqrt{2\pi}} e^{-\frac{2}{9}}} = \frac{0.8 \cdot e^{-\frac{1}{2}}}{0.8 \cdot e^{-\frac{1}{2}} + 0.2 \cdot e^{-\frac{2}{9}}}$$

$$\approx 0.752$$